Status	Finished
Started	Wednesday, 21 October 2026, 11:59 AM
Completed	Wednesday, 21 October 2026, 12:04 PM
Duration	5 mins 4 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 100
- c. 1000
- d. 10 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n ✓
- c. 1
- d. n/2

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By comparing each element sequentially
- c. By using a hash function
- d. By dividing the list length by two ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues in the left or right sublist ✓
- c. The search continues from the beginning
- d. The search stops

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There should be direct access to the middle element in any sublist
- b. Number values should only be present

  Number values should only be present

- c. There must be a mechanism to delete and/or insert elements in the list ✓
- d. The list must be sorted

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search starts over from the beginning
- b. The search backtracks to find duplicate elements
- c. The search continues until the end of the list
- d. The search stops immediately ✓

Question **7**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. At the beginning of the list
- c. In the right sublist ✓
- d. In the left sublist

Question **8**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Interpolation search
- b. Hash search ✗
- c. Binary search
- d. Linear search

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Dividing the list in half repeatedly
- b. Checking each element sequentially ✓
- c. Skipping every second element
- d. Sorting the list before searching

Question **10**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list is sorted
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It compares elements sequentially
- b. It can be applied only if the list is sorted ✓
- c. It works on unsorted lists
- d. It always starts from the beginning of the list

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- a. Searching ✓
- b. Discovering
- c. Finding
- d. Mining

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By using a hash function
- b. By comparing each element in the list sequentially ✓
- c. By dividing the list into halves
- d. By sorting the list first

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It can only be applied to large lists
- d. It is slower than linear search for small lists

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✕
- b. The middle element is chosen randomly
- c. The higher middle element is chosen as the middle element
- d. The search stops